AP PRECALCULUS · TOPIC 1.7

Rational Functions and End Behavior

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $f(x) = \dfrac{3x^2+1}{x^2-5}$. Which line is the horizontal asymptote of the graph of f?

A. $y = 0$ **B.** $y = 3$

C. $y = -\dfrac{1}{5}$ **D.** $x = \sqrt{5}$

2 Let $g(x) = \dfrac{2x+1}{x^2+4}$. What is $\lim\limits_{x\to\infty} g(x)$?

A. 2 **B.** $\dfrac{1}{4}$

C. ∞ **D.** 0

3 Let $h(x) = \dfrac{x^3-1}{2x^2+5}$. Which statement is true?

A. $h(x) \to \infty$ as $x \to \infty$ and $h(x) \to -\infty$ as $x \to -\infty$

B. $h(x) \to \infty$ as $x \to \infty$ and as $x \to -\infty$

C. $h(x) \to \dfrac{1}{2}$ as $x \to \infty$ and as $x \to -\infty$

D. $h(x) \to 0$ as $x \to \infty$ and as $x \to -\infty$

4 Let $r(x) = \dfrac{4-x^2}{2x^2+x}$. What is $\lim\limits_{x\to-\infty} r(x)$?

A. $\dfrac{1}{2}$ **B.** 2

C. $-\dfrac{1}{2}$ **D.** 4

5 Which function has a horizontal asymptote at $y = 2$?

A. $f(x) = \dfrac{2x+1}{x^2+1}$ **B.** $f(x) = \dfrac{x^2+2}{x-3}$

C. $f(x) = \dfrac{x+6}{3x^2}$ **D.** $f(x) = \dfrac{6x^2-x}{3x^2+5}$

6 A rational function $r(x) = \dfrac{p(x)}{q(x)}$ satisfies $\lim\limits_{x\to\infty} r(x) = 0$. Which must be true?

A. The degree of p equals the degree of q

B. The degree of p is less than the degree of q

C. The degree of p is greater than the degree of q

D. $p(x)$ has no real zeros

7 Let $r(x) = \dfrac{x}{x^2+1}$. Which statement is true?

A. The line $y = 0$ is a horizontal asymptote, and the graph never touches it

B. The line $y = 1$ is a horizontal asymptote

C. The line $y = 0$ is a horizontal asymptote, and the graph crosses it at $(0, 0)$

D. The graph has no horizontal asymptote

8 Let $f(x) = \dfrac{2x+3}{x-1}$. As $x \to \infty$, how does $f(x)$ approach its horizontal asymptote $y = 2$?

A. From above **B.** From below

C. It oscillates around $y = 2$ **D.** It never approaches $y = 2$

AP PRECALCULUS · TOPIC 1.7 · SECTION 2

Rational Functions and End Behavior

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

After chlorine is added to a city water reservoir, its concentration in mg/L is modeled by $C(t) = \dfrac{4t+3}{t^2+2}$, where $t \geq 0$ is the number of hours since the chlorine was added.

(a) Find $\lim_{t\to\infty} C(t)$. Interpret the value in context. DOK3 2 pt

(b) Find the greatest concentration and the time it occurs, to three decimal places. DOK3 2 pt

(c) After the peak, find the first time the concentration drops to 0.5 mg/L. Give your answer to three decimal places. DOK3 2 pt

Question 10 No calculator · 6 points

Let $f(x) = \dfrac{5x^2 - x}{2x^2 + 3}$ and $g(x) = \dfrac{x^3 + 1}{x^2 - 4}$.

(a) Write $\lim_{x\to\infty} f(x)$ and give the equation of the horizontal asymptote of f. DOK3 2 pt

(b) Describe the end behavior of g at both ends, using limit notation. DOK3 2 pt

(c) Does g have a horizontal asymptote? Justify using degrees. DOK3 2 pt